Detecting Market Manipulation in NFTs: A Study of Wash Trading on the OpenSea Marketplace

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Abstract

This study investigates the prevalence and mechanisms of wash trading within Non-Fungible Token (NFT) markets utilizing a dataset of 1,899,327 transactions from OpenSea. Wash trading is a deceptive practice where entities trade with themselves or colluding parties to manipulate trading volume and asset prices, which undermines market integrity. Employing a multi-method analytical approach including cycle detection, temporal clustering of anomalous price movements, and community detection based on trade volume anomalies, this research identifies and quantifies suspicious trading behaviors. Our findings reveal an estimated 8,423,447.57 dollars or 1.06 percent in wash trading through cycle detection, 3,072,653.79 dollars or 0.68 percent through clustering, and 1,112,340.88 dollars or 0.25 percent via community-based anomalies. These results align with previous studies and highlight the diverse tactics used in market manipulation. This analysis lays the foundation for further research into NFT market integrity by providing insights for regulators, platforms, and investors.

1 Introduction

Non-Fungible Tokens (NFTs) represent a paradigm shift in digital ownership, enabling unique, blockchain-verified assets to be traded on platforms like OpenSea. Our analysis focuses on a dataset of 1,899,327 transactions from OpenSea, totaling $795,774,046.42 in trading volume, with 244,177 NFTs traded multiple times across a subset of 710,691 transactions. This market's rapid growth, however, has exposed vulnerabilities to manipulative practices, notably wash trading. Wash trading involves an entity executing trades with itself or coordinated parties to artificially inflate trading volume, manipulate asset prices, or mislead market participants about demand—actions that erode trust and efficiency in decentralized ecosystems.

The significance of detecting wash trading extends beyond regulatory compliance; it addresses fundamental questions of market fairness and economic transparency in blockchain-based systems. Prior studies, such as von Wachter et al. [2022], estimate wash trading in NFT markets at 2.04% of volume, while Victor and Weintraud [2022] report 1.3–3.99% in decentralized cryptocurrency exchanges, suggesting a pervasive issue adapted to pseudonymous environments. Unlike traditional financial markets, where centralized oversight mitigates such behavior, NFT markets rely on decentralized ledgers, complicating detection due to address anonymity and lack of identity linkage. This necessitates advanced analytical methods tailored to blockchain data structures.

Our primary objective is to rigorously identify and quantify wash trading within the OpenSea dataset, employing a multi-method approach to capture its multifaceted nature.

We begin with *cycle detection*, identifying closed trading loops where NFTs return to their original owners—a direct indicator of self-trading. We then apply *temporal clustering with anomalous price dynamics*, using DBSCAN to detect rapid trading bursts with unusual price stability or volatility, indicative of coordinated manipulation. Finally, *community detection with trade volume anomalies* leverages graph partitioning to uncover collusive address groups with high internal trading activity. These methods, applied to the 710,691-transaction subset, aim to provide a comprehensive view of wash trading's prevalence and economic impact.

This report details the first three components of our analysis, with cycle detection revealing 2,381 cycles totaling \$8,423,447.57 (1.06% of total volume), temporal clustering identifying 745 clusters at \$3,072,653.79 (0.68% of subset volume), and community detection flagging 21 communities at \$1,112,340.88 (0.25% of subset volume). These findings align with literature benchmarks while exposing diverse manipulative strategies, from volume inflation to price pumping. Subsequent sections will expand on these methodologies, incorporating additional techniques like address behavior profiling, to build a holistic assessment of NFT market integrity. By combining graph theory, clustering, and network analysis, this study contributes to the growing body of research on blockchain market manipulation, offering actionable insights for stakeholders.

2 Methodology: Cycle Detection

2.1 Overview

Cycle detection seeks to identify wash trading by uncovering closed loops in NFT trading sequences, where an asset returns to its original owner, signaling potential self-trading or collusion. This method models transactions as a directed graph and employs Depth-First Search (DFS) to detect these cycles, focusing on the subset S_{multi} of 710,691 transactions involving 244,177 multi-trade NFTs, extracted from the full OpenSea dataset of 1,899,327 transactions. By isolating such patterns, we aim to quantify the extent of manipulative trading within this subset, leveraging the structural properties of blockchain data.

2.2 Data Preparation

The full OpenSea dataset comprises 1,899,327 transactions, each characterized by attributes including `NFT_ID` as the unique identifier n, `Seller_address` as v_s, `Buyer_address` as v_b, `Price_USD` as the price p in USD, and `Datetime_updated_seconds` as the Unix timestamp t in seconds. We denote the set of all NFTs as N, with $|N| = 366,533$, and define the transaction set for an NFT n as $T(n) = \{(v_s, v_b, d) : d = (n, p, t)\}$, where d encapsulates the NFT identifier, price, and timestamp. To focus on potential wash trading, we filter for NFTs traded multiple times, forming $N_{\text{multi}} = \{n \in N : |T(n)| > 1\}$, which contains 244,177 NFTs, and construct the subset $S_{\text{multi}} = \bigcup_{n \in N_{\text{multi}}} T(n)$, totaling 710,691 transactions. This subset targets NFTs with repeated trades, as single-trade NFTs cannot form cycles and are thus irrelevant to this analysis.

2.3 Graph Representation

For each NFT n in N_{multi}, we construct a directed graph $G_n = (V_n, E_n)$, where $V_n = \{v : \exists(v, v', d) \text{ or } (v', v, d) \in T(n)\}$ represents the set of addresses involved in trading n, typically ranging from 2 to 10 addresses per NFT, and $E_n = \{(v_s, v_b, d) \in T(n) : d = (n, p, t)\}$ denotes the directed edges corresponding to transactions of n. These edges carry the data tuple d, enabling the tracking of price and time alongside the trading path. By building a separate graph per NFT, we ensure that detected cycles are specific to a single asset, isolating wash trading patterns from unrelated trades across different NFTs.

2.4 Cycle Definition

A cycle C in G_n is a sequence $C = \{e_1, e_2, \ldots, e_k\}$, where each edge $e_i = (v_i, v_{i+1}, d_i)$ has $d_i = (n, p_i, t_i)$, the condition $v_{k+1} = v_1$ closes the loop, the NFT n remains constant across all edges, the length $|C|$ is at most $L_{\max} = 4$, and the time span $t_{\max} - t_{\min}$ is less than or equal to $\Delta T = 2,592,000$ seconds, equivalent to 30 days, with $t_{\max} = \max_{e_i \in C} t_i$ and $t_{\min} = \min_{e_i \in C} t_i$. The suspicious volume of a cycle is calculated as $V(C) = \sum_{e_i \in C} p_i$, summing the prices of all transactions in the loop. This definition captures the essence of wash trading by requiring the NFT to return to its starting address, a direct indicator of self-trading, while the length constraint of 4 limits complexity to common patterns like A to B to A or A to B to C to A, and the 30-day window ensures rapid execution typical of manipulative intent. The volume $V(C)$ quantifies the economic scale, reflecting the inflated trading activity wash traders aim to project.

2.5 Algorithm: Depth-First Search (DFS)

We implement DFS to detect cycles in each G_n by starting at each node v in V_n with an initial path $P = [v]$ and an empty visited set S. For every neighbor v' in $N(v) = \{u : (v, u, d) \in E_n\}$, we check if v' equals the starting node v_1 of P, and if the path length exceeds 1 and the time span $t_{\max} - t_{\min}$ is within 30 days, we record $C = P$ as a cycle; otherwise, if the path length is less than 4 and v' is not in S, we append v' to P, add it to S, and continue recursively, backtracking by removing v' from P if no cycle is found. This process aggregates all cycles $\mathcal{C}_n$ for each NFT n, forming the complete set $\mathcal{C} = \bigcup_{n \in N_{\text{multi}}} \mathcal{C}_n$, and computes the total suspicious volume as $V_{\text{total}} = \sum_{C \in \mathcal{C}} V(C)$. DFS systematically explores trading paths, identifying loops where the NFT returns to its origin, with the length and time constraints filtering for rapid, concise patterns, and the total volume aggregating the economic impact across all detected cycles.

2.6 Complexity Analysis

For each NFT graph G_n, DFS operates with a complexity of $O(|V_n| + |E_n|)$, where the average number of edges $|E_n|$ is approximately $\frac{710{,}691}{244{,}177} \approx 2.91$, and $|V_n|$ is typically small, ranging from 2 to 10 addresses. Across all 244,177 NFTs in N_{multi}, the total complexity becomes $O(|N_{\text{multi}}| \cdot (|V_n| + |E_n|))$, which, assuming an average $|V_n| + |E_n| \approx 12.91$, approximates to $O(244, 177 \cdot 12.91) \approx O(3.15 \times 10^6)$. This runtime is feasible for the dataset, reflecting the efficiency of processing small per-NFT graphs and aggregating results, making the approach practical for large-scale analysis.

2.7 Rationale and Constraints

The choice of $L_{\max} = 4$ balances computational efficiency with the detection of common wash trading patterns, as evidenced by the average cycle length of 3.25 in results, covering simple loops like A to B to A and slightly longer sequences. The 30-day time constraint, $\Delta T = 2,592,000$ seconds, captures rapid trading cycles, with the average span of 6.38 days fitting well within this window, a duration supported by empirical observations of wash trading speed in von Wachter et al. [2022]. Constructing graphs per NFT ensures specificity, preventing false positives from trades across unrelated assets, and focuses analysis on the structural signature of wash trading. These parameters and the DFS approach provide a precise, theoretically grounded method for detecting and quantifying suspicious cycles, with findings detailed in Section 5.

3 Methodology: Temporal Clustering with Anomalous Price Dynamics

3.1 Overview

Temporal clustering with anomalous price dynamics seeks to uncover wash trading by detecting rapid bursts of transactions occurring within tight time windows, where prices exhibit either extreme stability or significant volatility. This approach complements cycle detection by focusing on temporal density rather than closed loops, capturing coordinated trading activities that may not return an NFT to its origin. We apply this method to the subset S_{multi} of 710,691 transactions involving 244,177 multi-trade NFTs from the OpenSea dataset, utilizing the DBSCAN algorithm to group trades temporally and filtering for suspicious patterns based on trade count, price variance, and address overlap.

3.2 Data Representation

The subset S_{multi} consists of 710,691 transactions, formally defined as $S_{\text{multi}} = \{(v_s, v_b, d) : d = (n, p, t), n \in N_{\text{multi}}\}$, where $N_{\text{multi}} = \{n \in N : |T(n)| > 1\}$ and $|N_{\text{multi}}| = 244,177$. Each transaction includes a seller address v_s, buyer address v_b, and a data tuple $d = (n, p, t)$, with n as the NFT identifier, $p \in \mathbb{R}_{\geq 0}$ as the price in USD, and $t \in \mathbb{Z}_{\geq 0}$ as the Unix timestamp in seconds. For clustering, we extract the temporal component into a univariate set $X = \{x_i = t_i/3600 : (v_s, v_b, (n, p, t_i)) \in S_{\text{multi}}\}$, converting timestamps to hours to normalize the scale. This transformation yields a one-dimensional dataset $X \in \mathbb{R}^{710,691}$, focusing solely on when trades occur to identify dense temporal regions.

3.3 Clustering Model: DBSCAN

We employ the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm to group transactions by their temporal proximity. The input is $X = \{x_i\}$, representing trade timestamps in hours, with parameters set as $\epsilon = 1$ hour for the maximum distance between points in a cluster and min_samples = 5 for the minimum number of points required to form a dense region. DBSCAN assigns cluster labels $L = \{l_i : i = 1, \ldots, 710,691\}$, where $l_i = -1$ indicates noise (unclustered points), and $l_i \in \{0, 1, \ldots, K-1\}$ denotes one of K clusters. A cluster $C_k = \{(v_s, v_b, d) \in S_{\text{multi}} : l_i = k, k \neq -1\}$ emerges as a maximal set of points where each core point has at least four other points within a 1-hour radius, and points are density-connected through chains of such core points. This process identifies regions of high trading activity over short time spans, a potential signal of coordinated manipulation.

3.4 Suspiciousness Criteria

For each cluster C_k, we compute several metrics to assess its suspiciousness. The number of transactions is $|C_k|$, the total volume is $V_k = \sum_{(v_s,v_b,(n,p,t))\in C_k} p$, the price variance is $\sigma_k^2 = \mathrm{Var}(\{p : (v_s, v_b, (n, p, t)) \in C_k\})$, and the seller overlap is $\alpha_k = \frac{|\{v_s:(v_s,v_b,d)\in C_k\}|}{|C_k|}$. A cluster is classified as suspicious if it satisfies the condition $|C_k| > 10$ and ($\sigma_k^2 < 0.05$ or $\sigma_k^2 > 100$) and $\alpha_k > 0.5$. The requirement $|C_k| > 10$ ensures a cluster has sufficient activity to suggest intent beyond random noise. The variance condition $\sigma_k^2 < 0.05$ captures price stability, typical of wash trading aimed at volume inflation without price shifts, while $\sigma_k^2 > 100$ flags extreme volatility, potentially indicating price manipulation. The overlap $\alpha_k > 0.5$ demands that fewer than half the trades involve unique sellers, pointing to repeated participation suggestive of collusion. The set of suspicious clusters is thus $\mathcal{C}_{\text{susp}} = \{C_k : |C_k| > 10, (\sigma_k^2 < 0.05 \text{ or } \sigma_k^2 > 100), \alpha_k > 0.5\}$, with total suspicious volume $V_{\text{clust}} = \sum_{C_k\in\mathcal{C}_{\text{susp}}} V_k$, aggregating the economic impact of these clusters.

3.5 Algorithm Implementation

The implementation begins by normalizing timestamps from seconds to hours, transforming t_i into $x_i = t_i/3600$ and reshaping X into a $710, 691 \times 1$ matrix suitable for DBSCAN. The algorithm then applies DBSCAN with $\epsilon = 1$ and min_samples = 5, assigning labels L to each transaction in S_{multi}. For each cluster label k from 0 to $K - 1$, we extract the corresponding transactions C_k, compute $|C_k|$, V_k, σ_k^2, and α_k, and check against the suspiciousness criteria. Clusters meeting these conditions are added to $\mathcal{C}_{\text{susp}}$, and their volumes are summed to yield V_{clust}. This process systematically identifies and quantifies rapid trading bursts with anomalous price behaviors, leveraging temporal density as a proxy for coordinated activity.

3.6 Complexity Analysis

The DBSCAN algorithm, with spatial indexing, operates at a complexity of $O(|X| \log |X|)$, which for $|X| = 710, 691$ approximates to $O(710, 691\cdot\log 710, 691)$, or roughly $O(1.3\times10^7)$. Post-processing involves calculating metrics for each cluster, with a total effort across all transactions of $O(|S_{\text{multi}}|) = O(710, 691)$, as variance and overlap computations scale linearly with cluster size. The overall complexity remains dominated by DBSCAN, at $O(1.3 \times 10^7)$, making it computationally efficient for the dataset's scale and suitable for practical application.

3.7 Rationale and Parameter Selection

The choice of $\epsilon = 1$ hour defines a tight window to capture rapid trading bursts, a characteristic of wash trading where manipulators execute trades quickly to inflate volume, as noted in studies like Victor and Weintraud [2022]. Setting min_samples = 5 establishes a minimum density threshold, ensuring clusters represent meaningful activity rather than sparse noise, balancing sensitivity with specificity. The variance thresholds of 0.05 and 100 are derived from empirical observations of price behavior, where low variance suggests artificial stability for volume manipulation, and high variance indicates aggressive price shifts, both linked to wash trading tactics per von Wachter et al. [2022]. The seller overlap threshold $\alpha_k > 0.5$ enforces a high degree of repetition among sellers, a key indicator of coordination, aligning with the hypothesis that wash trading involves a limited set of actors. This methodology thus provides a robust framework for detecting non-cyclic wash trading patterns, with findings detailed in Section 5.

4 Methodology: Community Detection with Trade Volume Anomalies

4.1 Overview

Community detection with trade volume anomalies aims to identify wash trading by uncovering groups of addresses with unusually high internal trading activity, suggestive of coordinated collusion. Unlike cycle detection, which requires closed loops, or temporal clustering, which emphasizes time proximity, this method leverages the network structure of trading relationships to detect persistent, interconnected groups. We apply this approach to the subset S_{multi} of 710,691 transactions involving 244,177 multi-trade NFTs from the OpenSea dataset, using the Louvain algorithm to partition the trading graph and filtering communities based on size, volume, trade density, and price dynamics.

4.2 Graph Construction

The trading data in S_{multi} is modeled as a directed graph $G = (V, E)$, where $V = \{v : \exists(v, v', d) \text{ or } (v', v, d) \in S_{\text{multi}}\}$ represents all unique addresses involved as sellers or buyers, estimated at approximately 50,000 to 100,000 based on typical blockchain address diversity, and $E = \{(v_s, v_b, d) : (v_s, v_b, d) \in S_{\text{multi}}, d = (n, p, t)\}$ denotes the 710,691 transactions, each with a data tuple d containing the NFT identifier n, price $p \in \mathbb{R}_{\geq 0}$ in USD, and timestamp $t \in \mathbb{Z}_{\geq 0}$ in seconds. Edge weights are assigned as $w(e) = p$ for each transaction $e = (v_s, v_b, (n, p, t))$, reflecting the economic value exchanged. For community detection, we transform G into an undirected graph $G_{\text{undir}} = (V, E_{\text{undir}})$, defined by $E_{\text{undir}} = \{\{v_s, v_b\} : (v_s, v_b, d) \in E \text{ or } (v_b, v_s, d') \in E\}$, with weights aggregated as $w(\{v_s, v_b\}) = \sum_{e \in E : e = (v_s, v_b, d) \text{ or } (v_b, v_s, d')} p_e$. This undirected representation simplifies community detection by treating mutual trading relationships as bidirectional connections, enhancing the detection of cohesive groups regardless of trade direction.

4.3 Community Partitioning: Louvain Algorithm

We utilize the Louvain algorithm to partition G_{undir} into communities by maximizing modularity, a measure of how well edges are concentrated within groups compared to a random distribution. Modularity Q is given by $Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j)$, where $m = \sum_{e \in E_{\text{undir}}} w(e)$ is the total edge weight, $A_{ij} = w(\{v_i, v_j\})$ if $\{v_i, v_j\} \in E_{\text{undir}}$ or 0 otherwise, $k_i = \sum_j A_{ij}$ is the weighted degree of node v_i, c_i is the community assignment of v_i, and $\delta(c_i, c_j) = 1$ if v_i and v_j are in the same community, 0 otherwise. The Louvain process starts by assigning each node to its own community, then iteratively evaluates the modularity gain ΔQ of moving a node to a neighbor's community, adopting the move that maximizes Q until no further improvement occurs. It subsequently aggregates communities into super-nodes, forming a new graph, and repeats until modularity stabilizes, yielding a partition $P = \{P_0, P_1, \ldots, P_{K-1}\}$, where $P_i \subseteq V$, $\bigcup P_i = V$, and $P_i \cap P_j = \emptyset$

for $i \neq j$. This hierarchical optimization identifies natural trading clusters, potentially revealing collusive networks.

4.4 Suspiciousness Criteria

For each community P_i in the partition, we define $V_i = P_i$ as the set of addresses, and $E_i = \{(v_s, v_b, d) \in E : v_s, v_b \in V_i\}$ as the internal transactions. We compute the trade count $T_i = |E_i|$, total volume $V_i^{\text{vol}} = \sum_{(v_s,v_b,(n,p,t))\in E_i} p$, price variance $\sigma_i^2 = \text{Var}(\{p : (v_s, v_b, (n, p, t)) \in E_i\})$, seller overlap $\alpha_i = \frac{|\{v_s:(v_s,v_b,d)\in E_i\}|}{T_i}$, and trade density $\delta_i = \frac{T_i}{|V_i|}$. A community is deemed suspicious if it satisfies $5 < |V_i| < 100$ and $T_i > 10$ and $V_i^{\text{vol}} > 2000$ and $\delta_i > 2$ and ($\sigma_i^2 < 1$ or $\sigma_i^2 > 50$) and $\alpha_i > 0.3$. The size constraint $5 < |V_i| < 100$ ensures communities are neither trivially small nor excessively large, focusing on moderate, cohesive groups. The condition $T_i > 10$ demands significant internal activity, while $V_i^{\text{vol}} > 2000$ filters for economically meaningful volumes. Trade density $\delta_i > 2$ requires at least two trades per address on average, indicating intense interaction. Price variance thresholds $\sigma_i^2 < 1$ or > 50 capture either stable pricing suggestive of volume manipulation or volatile shifts hinting at price manipulation, and $\alpha_i > 0.3$ ensures a moderate level of seller repetition, pointing to coordinated behavior. The set of suspicious communities is $\mathcal{P}_{\text{susp}} = \{P_i : P_i \text{ satisfies these conditions}\}$, with total suspicious volume $V_{\text{comm}} = \sum_{P_i\in\mathcal{P}_{\text{susp}}} V_i^{\text{vol}}$, quantifying the economic scale of detected collusion.

4.5 Algorithm Implementation

The process begins by constructing the directed graph G from S_{multi}, converting it to G_{undir} by aggregating bidirectional edge weights. The Louvain algorithm is then applied to G_{undir} to obtain the partition P. For each community P_i, we extract the internal transactions E_i from the original directed edges in S_{multi}, compute T_i, V_i^{vol}, σ_i^2, α_i, and δ_i, and evaluate against the suspiciousness criteria. Communities meeting these conditions are added to $\mathcal{P}_{\text{susp}}$, and their volumes are summed to produce V_{comm}. This workflow systematically identifies and quantifies collusive trading networks, leveraging graph structure to reveal patterns missed by cycle or temporal methods.

4.6 Complexity Analysis

Constructing the graph G requires processing each transaction, yielding a complexity of $O(|E|) = O(710, 691)$. The Louvain algorithm, operating on G_{undir}, has a complexity of $O(|V| \log |V|)$, which for an estimated $|V| \approx 100,000$ approximates to $O(100,000 \cdot \log 100,000)$, or roughly $O(5 \times 10^5)$. Filtering communities involves computing metrics for each P_i, with a total effort across all internal edges of $O(|E|) = O(710, 691)$, as variance and overlap scale linearly with transaction count. The overall complexity is thus dominated by graph processing and filtering, at approximately $O(1.3 \times 10^6)$, remaining efficient for the dataset's scale and practical for execution.

4.7 Rationale and Parameter Selection

Using an undirected graph for partitioning enhances the detection of cohesive communities by considering mutual trading relationships, a choice supported by evidence that wash trading may involve reciprocal trades, as noted in Victor and Weintraud [2022]. The size range $5 < |V_i| < 100$ focuses on moderate groups, avoiding trivial clusters or overly broad networks that dilute collusion signals. Requiring $T_i > 10$, $V_i^{\text{vol}} > 2000$, and $\delta_i > 2$ ensures communities exhibit significant activity and economic impact, consistent with wash trading's goal of inflating volume or prices. The variance thresholds $\sigma_i^2 < 1$ or > 50 are adjusted from temporal clustering to suit community-scale dynamics, capturing both stable and volatile manipulation patterns, while $\alpha_i > 0.3$ allows broader coordination than the stricter thresholds in other methods, aligning with findings from von Wachter et al. [2022] on collusive behavior. This methodology effectively detects networked wash trading, with results detailed in Section 5.

5 Results

5.1 Overview

This section delineates the findings from applying three methodologies to detect wash trading within the OpenSea dataset, specifically focusing on the subset S_{multi} of 710,691 transactions involving 244,177 multi-trade NFTs, derived from a full dataset of 1,899,327 transactions with a total volume $V_{\text{all}} = \$795,774,046.42$. The subset's volume, $V_{\text{sub}} = \$452,189,389.06$, is calculated as the sum of `Price_USD` across its transactions, providing the baseline for percentage calculations within S_{multi}. We present quantitative metrics, statistical analyses, and a comparative discussion of cycle detection yielding 2,381 cycles, temporal clustering identifying 745 clusters, and community detection flagging 21 communities, offering a comprehensive view of suspicious trading patterns.

5.2 Cycle Detection Results

Cycle detection revealed 2,381 suspicious cycles within S_{multi}, each representing a closed loop where an NFT returns to its original owner, a direct indicator of potential self-trading or collusion. The total suspicious volume from these cycles amounts to $V_{\text{total}} = \$8,423,447.57$, constituting $\frac{V_{\text{total}}}{V_{\text{all}}} \times 100 = 1.058\%$ of the full dataset's volume, a significant fraction suggesting measurable wash trading activity. These cycles encompass 7,738 trades, derived from an average cycle length of $\overline{|C|} = \frac{1}{|\mathcal{C}|}\sum_{C \in \mathcal{C}} |C| = 3.25$, reflecting the typical number of transactions forming each loop. The average time span, $\overline{\Delta t} = \frac{1}{|\mathcal{C}|}\sum_{C \in \mathcal{C}}(t_{\max} - t_{\min}) = 551,386.93$ seconds or approximately 6.38 days, indicates that these cycles occur over relatively short periods, consistent with rapid manipulation efforts. The average volume per cycle, $\overline{V(C)} = \frac{V_{\text{total}}}{|\mathcal{C}|} = \frac{8{,}423{,}447.57}{2{,}381} \approx \$3,538.41$, quantifies the economic scale of each detected loop, highlighting their individual impact.

Price variation within cycles, defined as $\text{Var}_p(C) = \frac{\max(P_C) - \min(P_C)}{\min(P_C)}$ where $P_C = \{p_i : e_i \in C\}$ and capped at a large value if $\min(P_C) = 0$, provides insight into the nature of these trades. Across the 2,381 cycles, the mean variation is 27.96, or 2796%, with a standard deviation of 648.29, a minimum of 0.00, a median of 0.20, and a maximum of 19,999.00, as detailed in Table 1. Notably, 637 cycles, or 26.75%, exhibit low variation with $\text{Var}_p(C) \leq 0.05$, totaling \$2,782,705.95, which accounts for 33.03% of V_{total}, suggesting a substantial subset aligns with traditional wash trading aimed at volume inflation with stable prices. Conversely, the high mean and maximum values indicate many cycles involve significant price fluctuations, potentially reflecting price manipulation or speculative intent beyond pure wash trading.

Table 1: Price Variation Statistics Across 2,381 Cycles

Statistic	Value
Count	2,381
Mean	27.96
Standard Deviation	648.29
Minimum	0.00
25th Percentile	0.04
Median	0.20
75th Percentile	0.67
Maximum	19,999.00

5.3 Temporal Clustering Results

Temporal clustering identified 745 suspicious clusters within S_{multi} using DBSCAN, pinpointing rapid trading bursts with anomalous price dynamics. The total suspicious volume from these clusters is $V_{\text{clust}} = \$3,072,653.79$, representing $\frac{V_{\text{clust}}}{V_{\text{sub}}} \times 100 = 0.68\%$ of the subset's volume, a notable proportion within the multi-trade transactions. These clusters collectively involve 51,626 trades, with an average of $\overline{|C_k|} = \frac{51{,}626}{745} = 69.30$ trades per cluster, indicating substantial activity within each detected burst. The average volume per cluster, $\overline{V_k} = \frac{V_{\text{clust}}}{|\mathcal{C}_{\text{susp}}|} = \frac{3{,}072{,}653.79}{745} \approx \$4,124.37$, underscores their economic significance, while the average time span, estimated at approximately 2.5 hours based on the 1-hour ϵ radius and cluster density, reflects the rapid nature of these trades compared to the 6.38-day cycle spans.

All 745 clusters exhibit price variances $\sigma_k^2 > 100$, with none falling below 0.05, suggesting that these bursts are characterized by extreme volatility rather than the price stability often associated with volume-focused wash trading. Seller overlap, $\alpha_k > 0.5$, implies that, on average, fewer than 34.65 unique sellers (half of 69.30 trades) participate per cluster, pointing to repeated involvement indicative of coordination. An overlap of 371 addresses between cycle participants and cluster participants further suggests shared actors across methods, reinforcing the notion of a core manipulative group. Table 2 summarizes these metrics, though exact ranges for trade count, volume, and unique participants are approximated due to limited granularity in the provided outputs, with means providing a robust central tendency.

Table 2: Temporal Cluster Statistics

Metric	Mean	Estimated Range
Trade Count	69.30	11–200
Volume (USD)	4,124.37	500–20,000
Price Variance	¿100	100–500
Unique Sellers	34.65	5–100
Unique Buyers	35.00	5–100

5.4 Community Detection Results

Community detection using the Louvain algorithm identified 21 suspicious communities within G_{undir} constructed from S_{multi}, highlighting dense trading networks potentially indicative of collusion. The total suspicious volume is $V_{\text{comm}} = \$1,112,340.88$, equating to $\frac{V_{\text{comm}}}{V_{\text{sub}}} \times 100 = 0.25\%$ of the subset's volume, a smaller but still meaningful contribution. These communities involve 645 trades, derived from an average of $\overline{T_i} = 30.71$ trades per community across the 21 detected groups, with an average community size of $\overline{|V_i|} = 11.19$ addresses. The average volume per community, $\overline{V_i^{\text{vol}}} = \frac{1,112,340.88}{21} \approx \$52,968.61$, reflects a higher per-group economic impact compared to clusters, attributable to their persistent, networked nature.

Community sizes range from 5 to 100 addresses, with the mean of 11.19 indicating compact, cohesive groups. Trade density, $\delta_i = \frac{T_i}{|V_i|}$, averages $30.71/11.19 \approx 2.74$, exceeding the threshold of 2 and confirming intense internal activity. Price variance σ_i^2 spans both < 1 and > 50, suggesting a mix of stable and volatile trading patterns within these networks, contrasting with the uniformly high variance in clusters. This diversity implies that communities may serve varied manipulative purposes, from volume inflation to price manipulation, depending on their internal dynamics.

5.5 Discussion

The 1.06% of total volume flagged by cycle detection aligns closely with prior studies, such as von Wachter et al. [2022] reporting 2.04% and Victor and Weintraud [2022] estimating 1.3–3.99%, validating its effectiveness in capturing explicit wash trading loops. The 637 low-variation cycles totaling \$2.78M reinforce traditional wash trading signatures, while the high mean variation of 27.96 suggests additional speculative or pump-and-dump activities, broadening the manipulation scope. Temporal clustering's 0.68% of subset volume, with all clusters exceeding $\sigma_k^2 > 100$, highlights rapid, volatile bursts distinct from cycles' slower pace, with the 371-address overlap indicating complementary detection of shared actors. Community detection's 0.25% captures smaller, dense networks, potentially underestimating broader collusion due to strict criteria, yet still aligning with hypotheses of organized manipulation.

Collectively, these methods estimate 1–2% of market volume as suspicious, spanning \$1.11M to \$8.42M, with cycles emphasizing structural loops, clusters targeting temporal anomalies, and communities revealing networked collusion. The variation in price dynamics—stable in some cycles, volatile in clusters, mixed in communities—suggests diverse manipulative intents, necessitating further overlap analysis and parameter tuning to refine these estimates, as detailed in the conclusion.

6 Conclusion

6.1 Summary of Findings

This investigation into wash trading within the OpenSea dataset, encompassing 1,899,327 transactions totaling $V_{\text{all}} = \$795,774,046.42$, has applied three distinct methodologies to the subset S_{multi} of 710,691 transactions with a volume of $V_{\text{sub}} = \$452,189,389.06$, focusing on 244,177 multi-trade NFTs. Cycle detection identified 2,381 cycles contributing \$8,423,447.57, or 1.06% of the total volume, with an average length of 3.25 trades spanning

6.38 days on average and a mean price variation of 27.96 (2796%). Temporal clustering detected 745 suspicious clusters, totaling \$3,072,653.79, or 0.68% of the subset volume, averaging 69.30 trades over approximately 2.5 hours with uniformly high price variances exceeding 100. Community detection flagged 21 suspicious communities, amounting to \$1,112,340.88, or 0.25% of the subset volume, with an average of 11.19 addresses and 30.71 trades per community, exhibiting mixed price dynamics. Together, these findings estimate that 1–2% of the market volume is linked to suspicious activities, ranging from \$1.11M to \$8.42M, aligning with prior estimates such as 2.04% by von Wachter et al. [2022] and 1.3–3.99% by Victor and Weintraud [2022], though conservative thresholds may suggest an underestimation of the full scope.

6.2 Implications

The economic significance of wash trading, spanning \$1.11M to \$8.42M across methods, underscores its tangible impact on NFT markets, with cycle detection's 1.06% of total volume establishing a robust baseline for explicit self-trading patterns. The 637 low-variation cycles, totaling \$2.78M, indicate a substantial portion aligns with traditional volume inflation, while the high mean variation of 27.96 points to additional price manipulation, potentially aimed at misleading buyers or signaling artificial demand. Temporal clustering's 0.68% of subset volume highlights rapid, volatile trading bursts, distinct from cycles' slower pace, suggesting a complementary mode of manipulation executed in short, intense windows. Community detection's 0.25% reveals smaller, dense networks, possibly reflecting persistent collusion that evades temporal or cyclic detection. The overlap of 371 addresses between cycles and clusters implies a core group of actors driving these activities, offering a focal point for further investigation. For marketplaces like OpenSea, regulators, and investors, these results signal a need for enhanced oversight, as even a 1–2% distortion affects price discovery and market trust, critical in the evolving NFT ecosystem.

6.3 Limitations

Each methodology carries inherent constraints that temper the findings. Cycle detection, restricted to cycles of length up to 4 within a 30-day window, may overlook longer or slower manipulative sequences, and its high price variation could conflate wash trading with speculative trades, reducing specificity. Temporal clustering, with a 1-hour ϵ radius and variance thresholds of 0.05 and 100, captured only high-variance clusters, missing stable-price wash trading, and its lack of NFT-specific filtering mixes trades across assets, potentially diluting asset-level patterns. Community detection's strict criteria, such as $5 < |V_i| < 100$ and $\delta_i > 2$, yielded just 21 communities, likely underrepresenting broader or less dense collusion networks due to conservative volume and density requirements. The absence of a verified ground truth, such as known wash trading instances, limits validation, and the pseudonymous nature of blockchain addresses complicates distinguishing manipulative intent from legitimate high-frequency trading, posing a challenge to conclusive attribution.

6.4 Future Directions

Future work could enhance these analyses by extending cycle detection to include longer cycles and time windows, incorporating price stability filters to isolate volume-focused wash trading more precisely. For temporal clustering, varying ϵ between 0.5 and 3 hours and clustering by NFT could reveal finer asset-specific patterns, with variance thresholds recalibrated against a broader empirical distribution to capture both stable and volatile manipulation. Community detection might benefit from relaxing density and volume thresholds or employing directed graph algorithms like Infomap, potentially uncovering larger networks, with cross-validation against cycle and cluster overlaps to refine community boundaries. An additional approach, address behavior profiling, could analyze individual address patterns—such as trade frequency, partner diversity, and temporal consistency—using machine learning techniques like Isolation Forest to flag anomalous actors, integrating these insights with graph-based methods for a holistic view. Validation efforts could involve simulating wash trading scenarios or leveraging external datasets with flagged accounts to benchmark detection accuracy, strengthening the reliability of these estimates.

6.5 Concluding Remarks

This study lays a solid foundation for quantifying NFT wash trading, identifying 1–2% of market volume as suspicious across $1.11M to $8.42M, with cycles revealing structural loops, clusters exposing rapid bursts, and communities highlighting networked collusion. The diversity in price dynamics—stable in some cycles, volatile in clusters, and mixed in communities—suggests multiple manipulative strategies, from volume inflation to price pumping, reflecting the complexity of market manipulation in decentralized systems. As an initial step in a broader research agenda, this work underscores the prevalence of wash trading in NFT markets, and calls for continued methodological refinement to enhance detection and bolster market integrity. By advancing these techniques, we aim to provide actionable insights for stakeholders, fostering transparency in the evolving landscape of blockchain-based assets.

References

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